

Pharmacology: Drug Toxicities, Pharmacogenomics, and Drug Interactions

Practice Quiz

❖ Pharmacology bridges physiology, pathophysiology, and clinical practice.

CHALLENGE: Cover the options and try to answer the questions without looking at the choices.

Answers can be found by a SEARCH of the Notes handouts.

THE ANSWER KEY WILL BE POSTED FRIDAY NOV. 21.

PRELIMINARY RECALL QUESTIONS

How many risk factors for adverse drug reactions can you WRITE DOWN on a piece of paper or on e-paper?

Hint: You can find the answer in the PowerPoint.

Here they are. (There may be more.)

Age, especially the very young and the very old

State of the patient's health, nutrition, exercise

Pregnancy and the trimesters of the pregnancy

Drug exposure during organogenesis can cause teratogenic effects.

Some drugs have the potential for fetotoxicity during all trimesters.

Disease states: Effects are usually related to the severity of the disease and the patient's comorbidities

Genetic variabilities: Pharmacogenomics

Environmental exposures: smoking, vaping, contaminants in air, water, land

Concomitant interactions: prescriptions, herbal (botanical) products, over-the-counter medications, recreational drugs

NOW: Write in as many mechanisms of adverse drug effects that you think of for the following patient groups?

FILL IN

Older patients	
Pregnant patients	
Pediatric patients	

Patients with reduced organ function	
Patients on several concomitant medications	

Rationale:

Older patients	Altered pharmacokinetics and reduced organ function Polypharmacy Comorbidities
Pregnant patients	Fetus: teratogenicity / fetotoxicity Mom: altered PK, immune status, hypercoagulability
Pediatric patients	Neonates, especially premature infants, have immature drug clearance systems. These systems mature rapidly within months after birth (potentially longer in the premature infant). Infants and young children may have reduced or enhanced drug clearance – gestational age, young child's age
Patients with reduced function of organs of elimination	Reduced hepatic and renal function can reduce clearance. Reduced cardiac output can lead to reduced organ perfusion, leading to reduced hepatic and renal function.
Patients on several concomitant medications	Drug interactions leading to increased blood levels (potential toxicity) or decreased blood levels (potential subtherapeutic levels)

1. A 15-year-old female prescribed oral tetracycline for acne is cautioned not to take it together with dairy products, antacids, iron, calcium, or magnesium supplements, which directly bind to the drug.

What is the mechanism of this interaction?

- A. Inhibition of absorption*
- B. Inhibition of distribution
- C. Inhibition of metabolism
- D. Inhibition of excretion

Rationale: A. inhibition of absorption

The polyvalent metallic cations in antacids and nutritional supplements chelate the tetracyclines in the gastrointestinal (GI) tract, forming poorly absorbable metal-tetracycline complexes, which will be excreted in the feces.

Taking the drug together with the metallic cations infers that the complex formation will occur prior to absorption.

This effect is also classified as *chemical antagonism*.

2. A 67-year-old male presents to the internal medicine clinic with complaint of 8 to 10 watery stools per day for the past 7 days. He denies fever, blood in the stools, abdominal pain, and loss of appetite. His medications are a daily proton pump inhibitor for 6 months for self-diagnosed gastric reflux, daily Aleve (naproxen sodium) for self-diagnosed “arthritis pain”, and oral levofloxacin (antibiotic) for 14 days prescribed for a recurrent sinus infection, which was completed 10 days ago.

PMH: Appendectomy 3 months ago.

Lab: Stool samples are positive for *Clostridioides difficile* infection (formerly named *Clostridium difficile*).

How would this adverse drug event (ADE) be classified? (more than one answer)

- A. Dose related
- B. Duration related*
- C. Drug withdrawal
- D. Failure of therapy
- E. Not related to dose*
- F. Time-related, Acute
- G. Time-related, Delayed*
- H. Common and predictable*
- I. Rare and unpredictable

Suggestion: Highlight the clues in the stem.

Information to help you think through the question:

Microbiology: *C. difficile* is an anaerobic, gram-positive, spore-forming, toxin-producing bacillus that exists in spore and vegetative forms. It is highly transmissible by the fecal oral route and can be acquired in the hospital, where it can be cultured from hospital beds and other surfaces, as well as hands, clothing, and stethoscopes of health care workers. Either the spores or the vegetative form of the organism can be transmitted. *C. difficile* occupies the colon and is resistant to nearly all antibiotics.

Risk factors for infection:

- Antibiotics (especially broad-spectrum drugs like levofloxacin) disrupt the intestinal microbiome, allowing overgrowth of the non-susceptible microorganisms, such as *C. difficile*, which proliferates and expresses exotoxins. The toxins cause diarrhea and colitis.
 - *C. difficile*-associated diarrhea can occur days, weeks, even months, after completion of the antibiotic therapy.
 - Risk for *C. difficile* infection is increased with a longer duration of antibiotic therapy.
- Gastric acid suppression with proton pump inhibitors is associated with increased risk of *C. difficile* infection.
- NSAID use (naproxen sodium) can cause GI toxicity, which can sensitize GI epithelial cells to damage by pathogenic toxins.
- Older age (≥65 years of age) seems to promote increased frequency and severity of the infection.

Rationale: B, E, G, H

- **THE TAKEAWAY:** This ADE is a ***common and predictable, duration-related but not dose-related, delayed*** drug- disease adverse event.

Thinking it through:

A 67-year-old male presents to the clinic with diarrhea.

Symptoms: Copious watery stools for 7 days. Negative for fever, blood in the stools, abdominal pain, and loss of appetite.

PMH: Recent surgery and hospital stay, broad spectrum antibiotic therapy for 14 days, chronic proton pump inhibitor and NSAID usage

Diagnosis: Cause of the infection is confirmed by stool analyses.

Explanation:

- Gastrointestinal surgery carries a high risk of *C. difficile* infection
- Extended hospital stays increase exposure to *C. difficile* in the hospital environment
- Antibiotics disrupt beneficial bacteria in the gut microbiota, creating an environment where *C. difficile* can proliferate and cause infection
- Chronic acid suppression therapy neutralizes the stomach acid barrier. Stomach acid typically kills most ingested bacteria (fecal-oral route), including vegetative state of *C. difficile*, which allows *C. difficile* to proliferate.
- Chronic NSAID therapy can disrupt the epithelial barrier and act synergistically with *C. difficile* toxins to increase cell damage and permeability.
- Older adults are at increased risk due to age-related physiologic changes in the GI tract and decline in immune function, less diverse gut microbiome, increased health care exposure, more frequent use of antibiotics.

- ❖ **Epidemiology:** *Clostridioides difficile* is the most common nosocomial (hospital-/health care-acquired) cause of diarrhea. Most cases occur in people who have been hospitalized patients (hospital-acquired), but about 1/3 of cases are community-acquired. The hands of hospital personnel are important intermediaries.

The organism colonizes the large intestine of approximately 3% of the general population and up to 30% of hospitalized patients.

Note that most people are not colonized, which explains why most people who take antibiotics do not get pseudomembranous colitis caused by *C. difficile* toxins.

For your
information
if interested:
**No need to
memorize for
this class.**

Incorrect options:

- A. Dose-related = dose-dependent drug toxicities are related to the concentration of the drug in the body – the greater the dose, the greater the magnitude of effect.
C. difficile infection arises in susceptible individuals due to killing the beneficial organisms of the microbial flora, allowing *C. diff.* to overgrow and express toxins. It can occur with any antibiotic at any dose and is frequently a delayed effect.
- C. Drug withdrawal: *C. diff* infection can occur after drug withdrawal (discontinuation), but not as a result of it.
- D. Failure of therapy: The antibiotic eradicates the original infection (if it is not resistant). Therefore, the therapy is successful.
- F. Time-related, Acute: An adverse effect occurring rapidly from short-term exposure, even one dose
- G. Rare and unpredictable – “idiosyncratic” – adverse effects are not attributed to known pharmacologic or immune-related effects. They are very rare and cannot be predicted – unless a patient reports a previous reaction. *C. diff.* infection is common and predictable, which excludes rare and unpredictable as an option.

3. A pregnant 20-year-old person (she/her/hers) with a 2-year history of regular heroin abuse is admitted to the labor and delivery ward. She gives birth to a 6 lb, 2 oz boy, gestational age 40 weeks, by vaginal delivery. There are no signs of malformations. The baby exhibits hypertonia, tremors, high-pitched crying, poor sucking reflex, yawning, and autonomic instability (fever, temperature instability).

What is the most likely cause of the newborn’s signs/symptoms?

- A. Effect of Vitamin K shot given at birth
- B. Neonatal abstinence syndrome*
- C. Shock of coming out of warm, dark womb into a cold, bright room
- D. Teratogenic effects of the drug

Rationale: B. Neonatal abstinence syndrome

Neonatal abstinence syndrome due to abrupt opioid withdrawal is a predictable toxicity of opioid abuse by the mother. The stem states that no signs of malformations are present, indicating that the signs/symptoms are not related to a teratogenic effect of drug use.

4. A 26-year-old female presents to the family practice clinic with complaint of fever, severe left ear pain, and muffled hearing in the affected ear. She reports that she had cold symptoms with a runny, stuffy nose and cough for the last week. Antibiotic history is pertinent for penicillin for the treatment of strep throat 10 years ago, which she tolerated well. NKDA (no known drug allergies). Otoscopy shows bulging, erythematous tympanic membrane and reduced mobility of the tympanic membrane. An e-prescription is submitted for amoxicillin-clavulanate 875/125 mg by mouth twice daily (po BID).

Thirty minutes after taking the first dose, the patient develops a generalized itchy rash (urticaria) and facial swelling (angioedema).

What type of adverse drug reaction is described? Short answer

Rationale: Urticaria and angioedema (vasculitis involving deeper vessels) are signs of an immediate hypersensitivity reaction, Type I, IgE-mediated.



Hives (urticaria) on black skin

<https://www.healthline.com/health/skin/hives-on-black-skin>

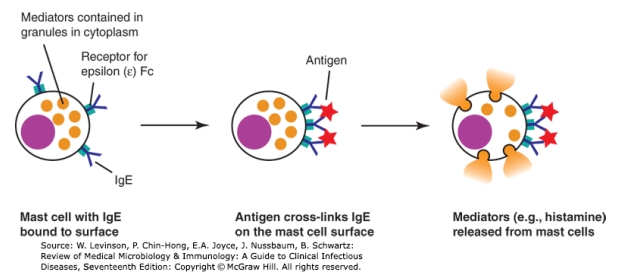


Figure 65-1 Medical Microbiology & Immunology: Antigen cross-links IgE on surface of mast cell. Mediators, e.g., histamine, are released from granules in the cytoplasm of the mast cell.

5. A 70-year-old female (pronouns she, her, hers) presents to the ED with intense, burning pain in her head, arms and legs, and signs/symptoms of a stroke. Medical history is significant for poorly controlled hypertension for ten years despite three antihypertensive medications and atrial fibrillation diagnosed 2 years ago for which she takes the anticoagulant warfarin.



Learning Radiology
<http://learningradiology.com/notes/general/notes/hemorrhagicstroke.htm>

Diagnostic studies:

- Findings on head CT without contrast are consistent with an acute intracerebral hemorrhage into the thalamus.
- Laboratory test shows elevated PT/INR, which is indicative of high blood levels of warfarin, which has a narrow therapeutic window.

Which of the following options describe the type of adverse drug effect? (more than one answer)

- A. Off-target effect
- B. On-target effect*
- C. Intended tissue
- D. Unintended tissue*
- E. Predictable effect*
- F. Unpredictable effect
- G. Dose-related*

Rationale: B, D, E, G

On-target effect: Warfarin's anticoagulant action occurs in all vascular tissues, including the brain.

Unintended tissue: The therapeutic use of the anticoagulant warfarin in patients with atrial fibrillation is prescribed to prevent thrombus formation in the left atrium (intended tissue), thus reducing the risk of stroke.

Anticoagulation in the vasculature of the brain (unintended tissue) can lead to stroke in the susceptible individual.

Predictable: Anticoagulants are administered for the purpose of reducing the tendency of the blood to clot. Bleeding is a predictable effect of anticoagulation.

Dose-dependent effect: The risk of intracranial bleeding increases with increased dose. Hypertension is an important risk factor for the development of intracerebral hemorrhage.

❖ **Practice point: Anticoagulation in patients with atrial fibrillation**

Atrial fibrillation causes uncoordinated electrical activations that impair left atrial contractility leading to reduced blood flow velocities.

Reduced blood flow leads to blood stasis in the left atrium, which can result in clot formation, potentially resulting in stroke.

Thrombus formation can be prevented by oral anticoagulant therapy.

Learning Radiology <http://learningradiology.com/notes/generalnotes/hemorrhagicstroke.htm>

Legend on the figure: Intracerebral hemorrhage, acute: Freshly extravasated whole blood. This bleed into the thalamus (thin white arrow) will be visible as increased density on non-enhanced CT scan of the brain due primarily to the protein in the blood (mostly hemoglobin). As the clot begins to form, the blood becomes denser for about 3 days because of

dehydration of the clot. After the 3rd day, the clot gradually decreases in density from the outside in and becomes invisible over the next several weeks.

UpToDate: Spontaneous intracerebral bleed: Pathogenesis, clinical features, and diagnosis

6. A 14-year-old male is given a diagnosis of seizure disorder and evaluated for treatment with the antiseizure drug, carbamazepine. It is the preferred drug for this patient based on his type of seizure, efficacy, side effect profile, and cost. The patient's parents are immigrants from Bangalore, India. Pharmacogenomics studies indicate that this patient may be at increased risk of developing a hypersensitivity reaction characterized by potentially life-threatening cutaneous reactions.

How might the risk be minimized?

- A. Screen for HLA genetic variants.*
- B. The reaction occurs rarely (up to 10% of patients) so standard dose should be prescribed.
- C. Start with low dose, titrate up, and monitor therapy.
- D. Select a different drug.

Rationale: A. Screen for HLA genetic variants

Learning point: Pharmacogenomics

Patients with HLA-B*15:02 allele are at increased risk of developing serious cutaneous adverse reactions (ie, those of Asian ancestry, including South Asians).

HLA-A*31:01, HLA-A*24:02 variants and others can increase the risk in a broader range of populations.

- If the patient tests positive for the HLA variant, an alternative drug should be selected.
- Note: A negative genetic test does not entirely rule out the risk of adverse drug reactions so patients should still be monitored.

Students, you do not need to memorize these details at this time.

It is the concept that is important:

- Patients vary in their responses to drugs – exceptional response, no benefit, and/or a range of adverse effects.
- Pharmacogenomic screening may help determine the effect of a medication in the individual patient.

7. Carbamazepine is an inducer of CYP3A4 metabolic enzymes and can *induce its own metabolism* (autoinduction).

i) Which of the following options is the most appropriate method for optimizing carbamazepine treatment for seizure control in the individual patient?

- A. Therapeutic plasma levels*
- B. Electroencephalograms
- C. Observation for seizure activity
- D. Signs and symptoms of skin rash

ii) What does “induce its own metabolism” mean in terms of drug effects over time?

Rationale: A. Therapeutic serum levels

Therapeutic drug monitoring can guide dosing. Carbamazepine assays and reference ranges are available.

Induction of CYP3A4 by carbamazepine increases the rate of metabolism of other CYP3A4 substrates and *its own metabolism* (autoinduction), thus reducing plasma levels of the drug, including its own plasma levels.

- **Practice point:** Closely monitor plasma carbamazepine levels during the first few months of therapy to determine if the blood levels are within the therapeutic range. If not, the dose can be increased according to published guidelines.

Therapeutic drug monitoring is useful when:

- there is an available, cost-effective drug assay for clinical use,
- there is a known relationship between dose and serum/plasma/blood concentration,
- there is an established target concentration
- the drug has a narrow therapeutic window (small therapeutic index),
- there is high patient variability in pharmacokinetic properties (drug liberation, absorption, distribution, metabolism, and excretion)
- there is a potential for severe, dose-related adverse effects

Concepts
to
remember

Incorrect answers:

- B. EEG traces seizure activity. It is not useful for directly determining therapeutic efficacy of the drug.
- C. Observation for seizure activity: The absence of seizure activity alone is not an indicator that the drug levels are within the therapeutic range. Conversely, the presence of seizure activity alone is not an indicator that the drug levels are outside of the therapeutic range.
- D. Skin rash is a hypersensitivity reaction related to an individual's predisposition for developing this adverse effect and not a predictor of effectiveness.

Patients should be monitored for signs/symptoms of skin rash. Mitigation of adverse effects is an essential component of safe drug therapy.

8. A 38-year-old woman complains to her doctor that she has been feeling sad, hopeless, and irritable for about 6 months. She has lost interest in everything. She relates that her family and co-workers have been urging her to get help. NKDA. After a full workup, she is given a diagnosis of major depressive disorder. An appointment is made with a mental health counselor and an electronic prescription (e-prescription) is submitted for an antidepressant that blocks the reuptake transporter of a specific neurotransmitter in the central nervous system and periphery. The antidepressant described in the vignette also binds the same type of receptors in the periphery and other types of receptors in both the periphery and the CNS.

i) Which one of the following options can be inferred about the drug's properties? (more than one answer)

- A. The drug has a high lipid:water partition coefficient.*
- B. The drug has high affinity for plasma proteins.
- C. The drug has high selectivity.
- D. The drug has high potency.
- E. The drug has high potential for side effects*

ii) What are the mechanisms of adverse effects?

Rationale: A. high lipid:water coefficient / E. high potential for side effects

Thought process for identifying the answers:

Properties:

- Major depressive disorder is a mental health disorder.
- Therefore, the drug target must be located in the central nervous system. It is.
- Therefore, the drug must cross the tight junctions of the blood-brain barrier to get to it.
- Lipophilic drugs are able to passively diffuse across the lipid bilayers of the endothelial cells and other cell types that form the blood-brain barrier.
- The antidepressant, therefore, must be lipophilic.
- The lipid:water partition coefficient characterizes the lipophilicity of substances. Therefore, this is one of the correct answer.

Mechanisms of adverse effects:

The drug's therapeutic target is the reuptake transporter of a specific neurotransmitter (a biogenic amine). This receptor type is widely distributed in the brain and the peripheral tissues.

The target tissue is the brain.

• **On-target adverse effects:**

Peripheral effects: Drug binding to reuptake transporters in the periphery can cause unwanted effects, such as diarrhea, and increased risk of bleeding.

CNS effects: Actions in areas of the CNS beyond those involved in mood and emotional responses. CNS side effects include insomnia, vivid dreams, restlessness, cognitive impairment, and sexual dysfunction

- **Off-target adverse effects:** Drug binding to other types of receptors in both the periphery and the CNS can cause unwanted effects, such as dry mouth, constipation, orthostatic hypotension / insomnia, vivid dreams, restlessness, cognitive impairment, and sexual dysfunction.

❖ **Pharmacology Pearl:**

Remember, pharmacodynamics and pharmacokinetics work together – There is no action without movement of drug through the body to the target; and when the drug is withdrawn, drug elimination will remove the action.

› FYI, Antidepressant mechanism of action: A reuptake transporter takes the neurotransmitter back into the presynaptic neuron out of the synapse, which ends their action on plasma membrane receptors.

Blocking reuptake causes the neurotransmitter to remain available in the synapse longer.

Incorrect options:

- B. Plasma protein binding: The fraction of the total amount of drug in plasma bound to plasma protein and not bound (unbound drug) determines the concentration of drug that enters the CNS (pharmacokinetics – drug distribution); only unbound drug moves between tissues.

This property does not relate to the mechanism of action or affinity of the drug for the receptor (pharmacodynamics).

- C. Specificity is the degree to which a drug gains access to and binds a select receptor type. A drug with high specificity interacts with a single type of receptor.

The vignette indicates that the drug binds its target AND other types of receptors both in the CNS and the periphery. It is not specific for the targeted receptor or targeted tissue.

- D. Potency refers to the amount of drug needed to produce a given effect. The antidepressant described in the vignette is effective. Therefore, it has affinity for its target. However, there is not enough information in the vignette to determine the drug's potency relative to some comparator.

9. A 48-year-old female presents to her PCP complaining of continuous, non-radiating muscle soreness in both thighs and calves for two days, which then worsened to crampy-type muscle pain (8 on a scale of 0-10) starting 4 hours ago. Three days ago, the patient went to the urgent care because of a “bad cough in my chest for one week but no fever”. The antibiotic clarithromycin was prescribed, and she states she has been taking it as directed. The patient's chronic medications are lisinopril for hypertension and high-intensity simvastatin for hypercholesterolemia. Review of her medical record shows complaints on routine visits of minor intermittent muscle aches and muscle cramping since starting the statin therapy. The patient is usually healthy, active, and happy with her life.

Labs: Serum creatine kinase (CK) level is slightly elevated. Urinalysis is negative for myoglobinuria, proteinuria, and hematuria.

If the signs and symptoms are related to her drug therapy, how could this adverse event have been avoided?

- A. Check for potential drug interactions before prescribing concomitant drugs.
- B. Do not prescribe antibiotics for viral illness (“a cough without fever”).
- C. If antibiotic therapy with a CYP inhibitor is required, hold the statin until antibiotic completed.

D. All options are correct.*

Information:

CK (also called creatine phosphokinase, CPK) is an indicator of skeletal muscle cell lysis.

Simvastatin is a CYP3A4 substrate.

High-intensity = high dose

Simvastatin can cause concentration-dependent muscle toxicity.

Clarithromycin is a CYP3A4 inhibitor.

Rationale: D. all options

The scenario describes the circumstance of a *predictable*, *unnecessary*, and *avoidable* drug interaction between the patient's statin and the antibiotic.

- Inhibition of CYP3A4 by the antibiotic inhibited the statin's metabolism, which led to elevated statin plasma levels, which led to muscle toxicity.
- Flags for potential drug interactions come up in the electronic medical records. These flags should not be ignored.
- Checking for potential drug interactions in advance can be done quickly by referring to a drug monograph or using a drug interactions app, such as UpToDate's Drug Interactions platform.

➤ **Practice point:** *Antibiotic therapy for a minor viral illness is inappropriate.* It will not eradicate the virus or shorten the illness, it unnecessarily exposes the patient to drug side effects, and it contributes to antibiotic resistance.

10. A 22-year-old male with schizophrenia resistant to first-line antipsychotic agents is initiated on treatment with clozapine, a highly effective drug for this indication but has a potential for causing serious adverse effects. The patient smokes two packs of cigarettes per day (10 pack-years). The clozapine dosage will be slowly titrated to the therapeutic plasma level. Because the patient is a smoker, a higher dose may be needed compared to nonsmokers. Monitoring signs and symptoms of serious toxicities including severe neutropenia, myocarditis and other cardiac anomalies, and more.

i) Which of the following metabolic enzymes is involved in this drug-smoking interaction?

- A. CYP1A2*
- B. CYP2B6
- C. CYP2C19
- D. CYP2D6
- E. CYP3A4

ii) Why might this patient need a higher dose to achieve serum levels within the therapeutic range than a patient who does not smoke?

Hint: the mechanism of the smoking-drug interaction

Rationale: A. CYP1A2

CYP1A2 induction by smoking (*ie*, tobacco and marijuana) increases the rate of metabolism of clozapine and other CYP1A2 substrates, reducing bioavailability, thus plasma levels, of the drug.

- **Practice point:** If a patient stops smoking during clozapine therapy, the patient may need a reduction in dose to ~70% of dose.

→ **Students are encouraged to know this smoking-CYP1A2 interaction for this course, for the Boards, and for clinical practice.**

CYP1A2 function:

CYP1A2 is a hepatic enzyme primarily involved in metabolism of various drugs, such as clozapine, and chemicals.

› FYI: Polycyclic aromatic hydrocarbons (PAHs) (figure) are planar aromatic compounds formed from incomplete combustion of organic matter, such as smoke, coal, gas, and some foods.

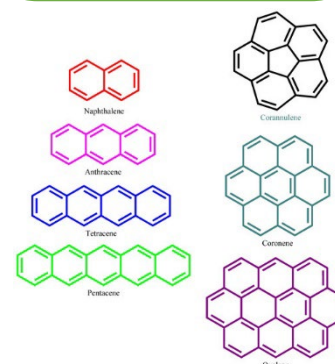
CYP1A2 metabolizes PAHs to **procarcinogens** by binding the nuclear receptor aryl hydrocarbon receptor, leading to increased CYP1A2 gene expression. Induction of CYP1A2 will increase the concentration of toxic reactive intermediate.

Smoking also induces other metabolic enzymes including CYP2B6, 2E1, 3A4, and UGT1A1.

The Takeaway:

The most significant induction by smoking is CYP1A2. Other metabolic enzymes are induced as well.

Enzyme induction by smoking can have clinical implications for a wide range of drugs.



11. The anti-hepatitis B virus (HBV) drug adefovir and the anti-HIV and anti-HBV drug tenofovir are both analogs of the nucleoside, adenosine. Here are the drug properties:

- They both are manufactured as prodrugs that facilitate their abilities to cross intestinal epithelial cell membranes by passive diffusion.
- They are both hydrolyzed to the active drug by different mechanisms.
- They are both converted by intracellular kinases to the phosphorylated form.
- They are both secreted as active drugs into the nephron via OATs.
- They both target the active site of the viral polymerase and compete for the same natural substrate.
- They are both associated with serious renal toxicity.

i) List all the potential mechanisms of this drug-drug interaction.

ii) What are the predictable clinical implications of combined therapy with adefovir and tenofovir?

Rationale:

The adefovir-tenofovir interaction involves elimination, mechanism of action, and toxicity:

- **Elimination:** Inhibition of organic anion transporters (OATs) in the proximal tubule can increase serum levels of the drugs.
- **Mechanism of action:** Tenofovir and adefovir bind near the same location within the active site of the HBV polymerase, and both compete for the natural substrate (dATP). A theoretical concern is competition for phosphorylation by the host kinases, direct evidence of competitive inhibition with reduce efficacy is lacking.
- **Toxicity – major concern:** Both adefovir and tenofovir are associated with proximal tubule dysfunction. Combining them could *amplify the nephrotoxic effects*.

Distribution within the body and metabolism are not sources of interference:

Both oral prodrugs cross enterocytes by passive diffusion, which is not saturable.

Both prodrugs are hydrolyzed to active drug, but by different routes.

Both drugs are excreted in active form – neither active drug undergoes hepatic metabolism

➤ **Practice point:** Adefovir-tenofovir *combination therapy should be avoided*. While efficacy could be reduced, the major concern is the potential for renal toxicity.

Tenofovir is effective in the treatment of both HIV and HBV. Adefovir is for HBV only. Some patients are coinfectd with both viruses. Tenofovir is a first-line anti-HIV drug. Adefovir should not be prescribed for its anti-HBV in this setting.

11. A 24-year-old woman presents to her PCP with a “bad sunburn” after being on beach for only one hour. The affected area is painful and interrupting her sleep. She admits that she ignored the doctor’s and pharmacist’s caution to avoid exposing herself to the sun while taking doxycycline. A bright red, confluent rash on her face, neck, upper chest, back, legs, and arms in sun-exposed areas is noted on physical examination. Welts and blistering are absent. She is advised to apply cold compresses to the affected area to relieve the symptoms and take acetaminophen for the pain.

What type of adverse drug reaction is described?

Please explain the mechanism.



UpToDate Graphic 86479 version 8.0

Rationale: Photosensitivity reaction / Phototoxicity

Tetracyclines, including doxycycline, are photosensitizers that become activated by UV light and form reactive oxygen species that cause direct cellular damage.

DEFINITIONS

Photosensitivity: an overarching term for any abnormal skin reaction caused by exposure to sunlight or artificial UV light.

Phototoxicity: A non-immune reaction that occurs when a photosensitizing chemical, such as a drug or metabolite, that absorbs UV light, creating free radicals or reactive oxygen species, resulting in *direct* cellular toxicity.

The reaction is dose-dependent.

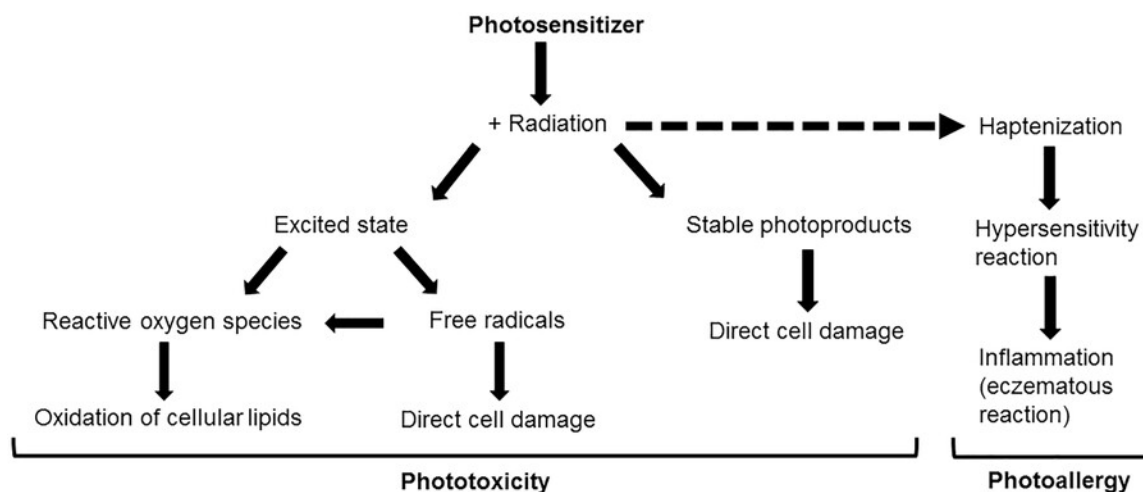
Photoallergic reaction: An immune-mediated, delayed hypersensitivity reaction that occurs when a photosensitizer present in the skin is chemically altered by UV light.

The photoproduct covalently binds skin proteins forming an antigenic chemical-protein complex that is recognized as foreign by Langerhans cells and other immune system components.

Eczematous, itchy lesions appear in 24 to 72 hours after sun exposure, can increase in severity, and may involve cell-mediated hypersensitivity.

The reaction is not dose-dependent.

Mechanisms:



Mechanism flow chart: Hofmann GA, Weber B. Drug-induced photosensitivity: culprit drugs, potential mechanisms and clinical consequences. J of German Society of Dermatology, Jan 2021; 19(1):19-29 Open Access Review Article <https://onlinelibrary.wiley.com/doi/full/10.1111/ddg.14314>

Summary table	Phototoxicity	Photoallergy
Mechanism	Direct cell damage	Immune-mediated hypersensitivity
Covalent binding with carrier protein	No	Yes
Cross-reaction with related agents	No	Yes
Incidence	High	Low
Dose-dependent	Yes (large amount required)	No
Onset	Minutes to hours	1 to 3 days
Prior sensitization required	No	Yes
Appearance	Exaggerated sunburn	Eczema-like dermatitis
Distribution	Exposed skin only	Exposed skin, may spread to unexposed skin
Pigmentary changes	Frequent	Unusual
UpToDate Graphic 50323, version 3.0; Appears in Photosensitivity disorders (photodermatoses)		

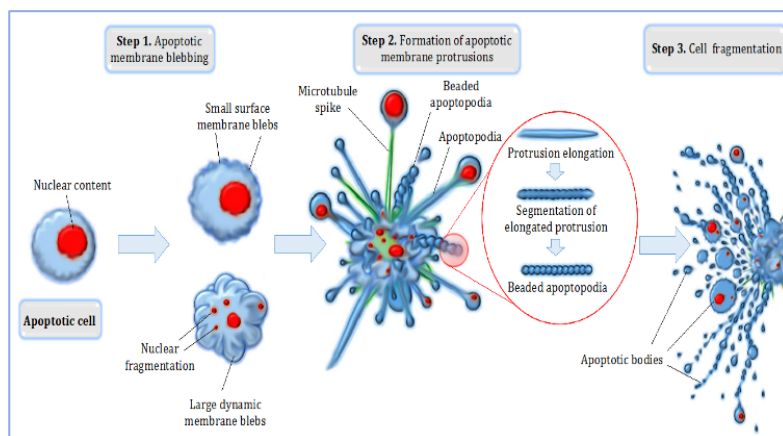
12. A patient is given a chemotherapy regimen for the treatment of breast cancer. The drugs directly damage the tumor cells' DNA and the DNA synthesis processes. The chemotherapy triggers the cell death pathway. The tumor cell fragments and dies without disrupting adjacent tissue.

What is the designation of the described process?

- A. Apoptosis*
- B. Autophagy
- C. Immune-mediated cell death
- D. Necrosis

Rationale:

Apoptosis is programmed cell death, a normal physiologic process for eliminating DNA-damaged, non-essential, or redundant cells. It is a series of coordinated, highly regulated biochemical events leading to cell shrinking, nuclear fragmentation, blebbing, and other cellular changes. This mechanism induces minimal inflammation and disruption of adjacent tissue.



Sources: Figure: Aaron Smith, Michael AF Parkes, Georgia K Atkin-Smith, Rochelle Tixeira, Ivan KH Poon - [Wikiversity:Draft:WikiJournal of Medicine/Cell disassembly during cell death](#)

13. An 86-year-old man with valvular heart disease taking the anticoagulant warfarin (a vitamin K antagonist) presents to his cardiologist because of a laboratory finding of significantly elevated PT/INR of 14 (normal therapeutic range 2-3). He reports that he has not had nose bleeds or bleeding gums. The patient was hospitalized for management of intracerebral bleeding 8 months ago. Bruises are noted on bilateral upper arms. Signs and symptoms of serious bleeding are absent. Given the patient's high bleeding risk, he is given the reversal agent, oral vitamin K, and directed to stop taking the warfarin for now. He is to return to the office tomorrow for PT/INR testing.

What is the DIRECT action of vitamin K in this context?

- A. Outcompeting warfarin at vitamin K reductase*
- B. Preventing serious bleeding event
- C. Supplementing dietary intake of vitamin K

Information on the drug mechanism: Warfarin is a competitive antagonist of the enzyme necessary for activating vitamin K, which is necessary for activating the synthesis of hepatic coagulation factors.

Rationale: A. Outcompeting warfarin at the enzyme, vitamin K reductase

Clue: "Competitive antagonist" indicates that the drug will outcompete the substrate at the enzyme, thereby antagonizing the effect of the enzyme-substrate interaction.

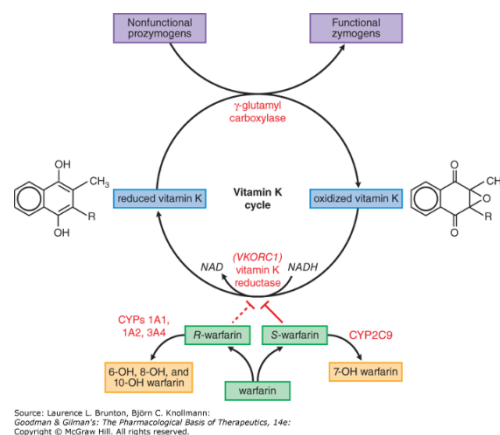
Mechanism of the drug-vitamin K interaction:

Vitamin K reductase catalyzes the conversion of inactive oxidized vitamin K to active reduced vitamin K.

Warfarin is a competitive antagonist of vitamin K reductase, which prevents the substrate, oxidized vitamin K, from accessing the catalytic site.

Therefore, increasing the concentration of vitamin K will outcompete warfarin for the catalytic site on the enzyme, allowing increased concentrations of reduced vitamin K necessary for the synthesis of hepatic coagulation factors.

This scenario is an example of a **beneficial interaction**.



Goodman & Gilman Figure 36-6

Incorrect options:

The **therapeutic goal** (not the action) of the vitamin K administration is to prevent serious bleeding. That is, the indirect effect of therapy is prevention of serious bleeding.

Regarding vitamin K in food: Vitamin K in foods can reduce the anticoagulant effects of warfarin. Consumption of vitamin K-containing foods should be consistent during warfarin therapy.

Sources: Dark green leafy vegetables, broccoli, cabbage, asparagus, and several other foods.

Legend on Figure 36-6: The vitamin K cycle and mechanism of action of warfarin. In the racemic mixture of S- and R-enantiomers, S-warfarin is more active. By blocking VKOR encoded by the VKORC1 gene, warfarin inhibits the conversion of oxidized vitamin K epoxide to its reduced form, vitamin K hydroquinone. This inhibits vitamin K–dependent γ -carboxylation of factors II, VII, IX, and X because reduced vitamin K serves as a cofactor for a γ -glutamyl carboxylase that catalyzes the γ -carboxylation process whereby prozymogens are converted to zymogens capable of binding Ca^{2+} and interacting with anionic phospholipids. S-Warfarin is metabolized by CYP2C9; common genetic polymorphisms in this enzyme increase warfarin metabolism. Polymorphisms in the VKORC1 increase the susceptibility of the enzyme to warfarin-induced inhibition. Thus, patients expressing polymorphisms in these two enzymes require reduction of warfarin dosage (see Table 36–2).

14. A 16-year-old female presenting to the women's health clinic believes she is pregnant and has been hiding it from her parents. She has not received prenatal care. She takes valproic acid for seizure disorder. Ultrasound images are consistent with 20 weeks gestation. A neural tube defect is noted.

What two terms describe this adverse drug event?

Rationale:

Fetotoxicity: Harmful effects on a developing fetus at any point in the pregnancy resulting from maternal exposure to a medication or other substance toxic to the fetus

Teratogenicity: Structural or functional abnormalities in a developing embryo or fetus during organogenesis caused by maternal exposure to a medication or other substance

Teratogens are agents that cause the congenital malformations, such as neural tube defects. NTDs are a form of fetotoxicity.

The dose and duration of the embryo's or the fetus' exposure to the teratogen are important.

The congenital malformation described in the vignette is a ***predictable*** adverse fetal effect of the valproate based on animal reproduction studies and surveillance data of teratogenic effects in children born to mothers taking valproic acid.

Birth defects can be caused by genetic abnormalities and/or environmental exposures. The underlying etiology is often unknown.

Determination of teratogenicity:

FYI: It can be difficult to determine if a substance is teratogenic.

The FDA requires that all prescription drugs be tested in animal models. These results may not always apply to humans. Human data regarding the association of drugs (and other chemicals) with birth defects comes from case studies and post-marketing surveillance through drug registries, which rely on the participation of clinicians and pregnant patients.

15. A 58-year-old woman with resistant hypertension is given add-on therapy with a vasodilator called hydralazine. Four weeks later she presents to the cardiologist with fever, malar rash (photo), myalgia, and arthralgia. She states she has been taking all her blood pressure medicines exactly as directed and that she is not taking any other drugs or recreational substances. There are no drug interactions with her antihypertensive medications. Suspecting a lupus-like syndrome related to elevated plasma drug concentrations due to deficient activity of the enzyme that inactivates hydralazine by acetylation, the drug is discontinued and her antihypertensive therapy is adjusted. The symptoms/signs resolve.



Picture: Journal of Clinical Pharmacy and Therapeutics
<https://onlinelibrary.wiley.com/doi/10.1111/jcpt.12506>

i) If the adverse drug effect is due to deficient activity of the drug metabolizing enzyme, which one of the following options is the term that describes this patient's pharmacogenetic phenotype?

- A. Poor metabolizer*
- B. Intermediate metabolizer
- C. Extensive metabolizer
- D. Ultra-rapid metabolizer

ii) Which drug metabolizing enzyme is involved?

- A. NAT2*
- B. CYP2D6
- C. G6PD
- D. VKORC1

Fill the blanks (Hint: Handout p15-16)

Metabolizer Type	Enzyme Activity	Clinical Impact	
		Standard Drug	Prodrug
Poor metabolizer PM			
Intermediate metabolizer IM			
Extensive metabolizer EM			
Ultrarapid metabolizer UM			

Rationale: A. Poor metabolizer / A. NAT2 – “A” is for acetylation

Clues:

- Presentation: Dose/concentration-related drug-induced adverse effect
 Students: A fundamental principle of pharmacology is that the dose / plasma concentration is related to the magnitude of effect. This includes the potential for and the magnitude of side effects.

As with anything, the rule has exceptions but it is almost always true.

- Patient is taking the prescribed dose of drug.
Students: You can assume standard dosing and that the patient is not taking more drug than is prescribed. If she were, it would be stated in the vignette.
- Patient is not taking other drugs that might interact.
Students: If the veracity of the patient's statement were in doubt, it would be stated in the vignette.
- The drug is inactivated by acetylation.
Acetylation is catalyzed by acetyltransferase, a phase 2 reaction. NAT2 is the primary isoform for biotransforming substrates in the liver.
NAT2 genetic variations lead to variations in how individuals metabolize substrates – drugs and carcinogens.

Thinking it through:

What mechanisms lead to increased plasma concentration?

- High dose: This cause has been ruled out.
- Inhibition of metabolism by another drug:
This cause is unlikely because the drugs in the antihypertensive regimen do not have mechanism/transport-related interactions and the patient states she is not taking other drugs.
- Reduced activity of the metabolic enzyme:

The giveaway:

“Suspecting a lupus-like syndrome related to elevated plasma drug concentrations due to deficient activity of the enzyme that inactivates hydralazine by acetylation”

Therefore, the patient must be a “poor metabolizer”, meaning the patient has genetic variants that lead to reduced or absent function of the drug metabolizing enzyme – acetyltransferase type 2 (NAT2) in this example.

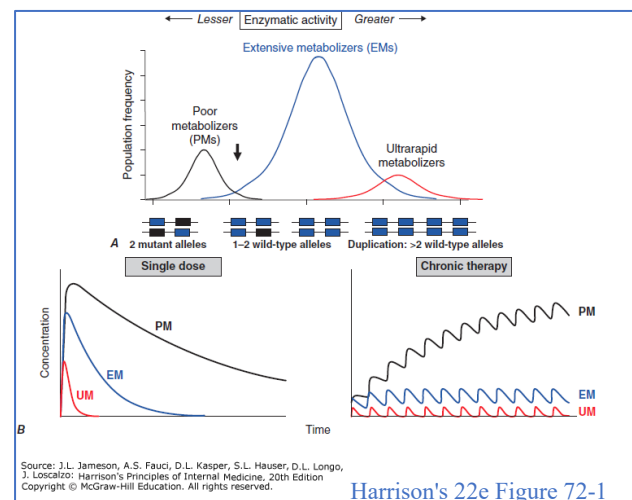
Learning points: (summarized in the table below)

Pharmacogenetic variability in the genes expressing the drug metabolizing enzymes

***Poor metabolizers (PM phenotype):** Individuals with two alleles (variants) encoding for nonfunctional protein

- ❖ Prodrugs, which require metabolic activation, may not be converted to the active compound in individuals with reduced-function haplotypes (poor metabolizers), leading to subtherapeutic levels and reduced response.

***Intermediate metabolizers, IM:** Individuals with one functional allele; these individuals may or may not be distinguishable from extensive metabolizers (two functional alleles)



Harrison's 22e Figure 72-1

***Extensive metabolizers, EM:** Individuals with two functional alleles

***Ultra-rapid metabolizers, UM:** Very high enzymatic activity, due to gene duplication or multi-duplication.

Harrison's Figure 64-1: CYP2D6 Metabolizers (PM, EM, UM)

Panel A: Distribution of CYP2D6 metabolic activity across a population. The heavy arrow indicates a single antimode, separating poor metabolizer subjects (PMs, black) with two loss-of-function CYP2D6 alleles (black), indicated by the intron-exon structure below the chart. Individuals with one or two functional alleles are grouped together as extensive metabolizers (Ems, blue). Also shown are ultra-rapid metabolizers (Ums, red), with 2-12 functional copies of the gene, displaying the greatest enzyme activity.

Panel B, left. The simulations show the predicted effects of CYP2D6 genotype on disposition of a substrate drug. With a single dose, there is an inverse “gene-dose” relationship between the number of active alleles and the areas under the time-concentration curves (smallest in UM subjects; highest in PM subjects); this indicates that clearance is greatest in UM subjects. In addition, elimination half-life is longest in PM subjects.

Panel B, right. The single dose differences are exaggerated during chronic therapy: steady state concentration is much higher in PM subjects (decreased clearance), as is the time required to achieve steady state (longer elimination half-life)

→ THE TAKEAWAY (know this)

Metabolizer Type	Enzyme Activity	Clinical Impact	
		Standard Drug	Prodrug
Poor metabolizer PM	Little to none	Drug accumulation ↑ Risk of toxicity	Prodrug may not be converted to active compound
Intermediate metabolizer IM	Reduced	Reduced activity May ↑ risk of side effects	Prodrug may have reduced response
Extensive metabolizer EM	Normal	Normal Typically effective / tolerable	Normal rate of conversion Typically effective
Ultrarapid metabolizer UM	Increased	Rapid clearance May ↓ efficacy	High levels of active compound Potentially dangerous toxicity

PMs: Individuals with two alleles (variants) encoding for nonfunctional drug metabolizing enzyme.

IMs: Individuals with one functional allele and one nonfunctional allele. Metabolism is slower than EMs but not as slow as PMs. Drug levels may be too high at standard doses.

EMs: Individual with two functional alleles. Normal metabolizers

UMs: Individuals with increased enzymatic activity, such as gene duplications. Very rapid metabolism

➤ **Practice points:** Knowing how a patient will respond to certain drugs helps clinicians tailor treatment plans and minimize risk.

Pharmacogenetic testing is appropriate when there is a known and clinically significant gene-drug interaction, drugs with a narrow therapeutic window or high risk of adverse effects, or a history of treatment failure.

16. A healthy 30-year-old female presenting to the obstetrician is pleased to find out that she is 8 weeks pregnant. She asks how the doctor knows if various over-the-counter and prescription medications she might need during pregnancy are safe for her unborn baby. The doctor explains the FDA has issued guidance for the use of medications in pregnancy and breastfeeding to help patients understand the potential risks and benefits of medication use.

i) What is the name of this set of regulations on drug and biological product labels?

(Hint: PLLR)

ii) What three main categories of information are provided in this FDA guidance document?

Rationale:

i) Pregnancy and Lactation Labeling Rule

ii) PLLR provides information on data, risk, and management

- **Data** on potential risks in the mother and fetus
- **Risk summary** on potential adverse effects to the developing fetus throughout the pregnancy and pharmacokinetic parameters altered during pregnancy.
- **Clinical considerations** are provided about initiation, continuation, and withdrawal of drug treatment during pregnancy

The **goal of the PLLR** is to provide health care providers with information needed to make informed decisions about prescribing medications to pregnant and breastfeeding individuals, and to help patients understand the potential risks and benefits of medication use.

17. A 74-year-old man with a history of stable angina is celebrating his birthday. His buddy gives him a “little blue pill” (Viagra) so he can “have fun tonight”. The man takes the Viagra (sildenafil) and collapses on the floor (from hypotension). His wife calls an ambulance. The patient’s chronic anti-angina medications are a long-acting beta blocker and a long-acting nitrate. What is the mechanism of this drug-drug interaction that resulted in the adverse drug event?

- A. Additive*
- B. Competitive
- C. Functional
- D. Mutualistic

Information:

Beta-blockers slow heart rate and reduce the force of cardiac contraction, which reduce the work of the heart and lower blood pressure.

Nitrates cause dilation of veins and arteries, which reduces the work of the heart and lower blood pressure.

PDE-5 inhibitors like Viagra indirectly sustain levels of nitric oxide in vascular smooth muscle by preventing breakdown of nitric oxide.

Rationale: A. Additive

This scenario illustrates an additive drug-drug interaction producing a harmful adverse drug event – profound hypotension.

Concurrent administration of the two drugs is *contraindicated*.

FYI: Some sources call the drug interaction additive and some call it synergistic.

Bottom line, the combination causes dangerous hypotension.

Patients prescribed nitroglycerin should be cautioned about the dangers of taking it with any PDE5 inhibitor.

Patients prescribed a PDE5 inhibitor should be cautioned about the dangers of taking it with other drugs that cause hypotension.

Mechanism:

- Nitroglycerin forms NO → activates cGMP → vascular smooth muscle relaxation → ↓BP
- Sildenafil inhibits cGMP degradation → ↑cGMP levels → vascular smooth muscle relaxation → ↓BP

PDE5 inhibitors erectile dysfunction mechanism:

Physiology: During penile stimulation nitric oxide is released in the corpus cavernosum. Nitric oxide activates guanylyl cyclase, which increases levels of cGMP, producing smooth muscle relaxation and inflow of blood into the corpus cavernosum.

Pharmacology: PDE5 inhibitors enhance the action of nitric oxide by inhibiting degradation of cGMP in the corpus cavernosum. It has no effect on in the absence of sexual stimulation taken at recommended doses.

Incorrect options:

- B. Competitive antagonism can occur when one drug competing for the same receptor as another drug interferes with the actions of the other drug.
- C. Mutualistic drug interaction can occur when two drugs acting on different but complementary pathways act in a beneficial and cooperative way to achieve therapeutic effect that neither could achieve effectively alone.
- D. Functional drug interaction can occur when two drugs acting on different pathways produce opposite, antagonistic effects.

A beneficial functional interaction was not covered in the lecture.

It makes sense that drugs acting on different pathways could have positive effects, such as enhancing absorption or efficacy.

If you defined it this way, then you are correct and there are two answers.

18. A 64-year-old woman is prescribed an oral anticoagulant called apixaban for treatment of atrial fibrillation. On questioning she reports having a glass of grapefruit juice every day for breakfast and dinner. She is cautioned that grapefruit juice can increase the plasma levels of apixaban, which could lead to serious bleeding, including bleeding in the brain. The patient agrees to switch to orange juice.

What enzymes metabolize apixaban?

Rationale: CYP3A4

Grapefruit juice contains a furanocoumarin called bergamottin, which is metabolized by CYP3A4 to reactive intermediates that bond covalently (irreversibly) to the active site of CYP3A4 enzymes.

Grapefruit juice consumption inhibits CYP3A4 activity in intestinal epithelial cells and in hepatocytes.

Drugs with documented increase in oral bioavailability due to the grapefruit juice-drug interaction are those with low and intermediate intrinsic bioavailability because the lower the intrinsic bioavailability, the greater the possible increase in systemic concentration when CYP3A4 is inhibited compared to drug which have high intrinsic bioavailability (low hepatic extraction).

For example,

If a drug has an intrinsic bioavailability of 10% and the bioavailability increases because of CYP3A4 inhibition to 50%, that is a 5-times increase in bioavailability.

If a drug has an intrinsic bioavailability of 98%, inhibition of an enzyme might increase the bioavailability to 99% - a very small difference.

Source:

Bailey DG, Dresser G, Arnold JM. Grapefruit-medication interactions: forbidden fruit or avoidable consequences? CMAJ. 2013 Mar 5;185(4):309-16. doi: 10.1503/cmaj.120951. Epub 2012 Nov 26. PMID: 23184849; PMCID: PMC3589309. Accessed Aug 3, 2023

[https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3589309/#:~:text=The%20chemicals%20in%20grapefruit%20involved%20in%20this%20interaction%20are%20the%20furanocoumarins.&text=Furanocoumarins%20are%20metabolized%20by%20CYP3A4,\(mechanism%2Dbased%20inhibition\)](https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3589309/#:~:text=The%20chemicals%20in%20grapefruit%20involved%20in%20this%20interaction%20are%20the%20furanocoumarins.&text=Furanocoumarins%20are%20metabolized%20by%20CYP3A4,(mechanism%2Dbased%20inhibition))

Sevrioukova IF. Structural Insights into the Interaction of Cytochrome P450 3A4 with Suicide Substrates: Mibefradil, Azamulin and 6',7'-Dihydroxybergamottin. Int J Mol Sci. 2019 Aug 30;20(17):4245. doi: 10.3390/ijms20174245. PMID: 31480231; PMCID: PMC6747129.

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6747129/>

19. A 54-year-old woman with a history of tophaceous gout manages hyperuricemia with probenecid, a competitive inhibitor of uric acid transport in the proximal tubule, which increases uric acid excretion. Probenecid also inhibits several isoforms in the same super family of proteins, which reduces excretion of drugs, such as penicillin, cephalosporins, ibuprofen, and many others.

From the description, you can deduce what protein is inhibited by probenecid.

What is it?

- A. CYP3A4
- B. Na^+, K^+ -ATPase
- C. Organic anion transporter*
- D. Organic cation transporter
- E. P-glycoprotein

Definitions: tophaceous – rough, stony; tophaceous gout – uric acid crystals (tophi) deposit in soft tissues and form hard nodules.

Rationale: D. organic anion transporters

Thinking it through

- Probenecid inhibits uric acid transport.
- Uric acid is a weak acid that dissociates into an organic anion, urate.
- Organic anions are substrates of organic anion transporters (OATs).
- Probenecid inhibits the urate transporter and additional organic acid transporters.
- Therefore, probenecid use can result in increased systemic concentrations of those substrate drugs.

20. JB is a 35-year-old man with a 10-year history of HIV infection well managed with antiretroviral therapy. He is a health care worker in a group home where he has been in close contact with a resident who has been diagnosed with meningococcal meningitis, which is caused by *Neisseria meningitidis*. One of the antibiotic options for chemoprophylaxis is a strong CYP3A4 inducer that would decrease the efficacy of his anti-HIV therapy.

What is the drug given to prevent the CNS infection?

- A. Ceftriaxone
- B. Ciprofloxacin
- C. Erythromycin
- D. Rifampin*
- E. Tetracycline

Rationale: Rifampin is a strong inducer of several CYPs and transporters.

Thinking it through:

- Rifampin is one of the starred drugs that is an important CYP3A4 (and P-gp) inducer.
- Erythromycin is a starred drug that is an important CYP3A4 (and P-gp) inhibitor.
- The drugs in the other options have not been specifically described or starred in the series of lectures to this point so you can immediately discard them as correct options.

In summary, even without being familiar with the drugs in all the options, you are able to recognize the correct option.

FYI: The drugs in all options are antibiotics. You will be learning about antibiotics in the FOM course.

Looking ahead: Ceftriaxone (250 mg IM x1) and ciprofloxacin (500 mg po x1, if low rates of resistance in the region) are alternatives for the prophylaxis of meningococcal meningitis and would be a better choice for the health care worker in the case vignette.

21. A 38-year-old female presents to the ED with swelling, pain, redness, and warmth in her left lower extremity for three days. She describes the pain as a deep soreness (7 on a scale of 1 to 10) that increases with physical activity. She states that she tried cold compresses, elevation, ibuprofen, and acetaminophen. Nothing made it better or worse. The patient has a 10-year history of bipolar disorder stabilized on carbamazepine and lithium. She takes no other prescription drugs. She states that, as an insurance agent, she sits all day and doesn't get any exercise. She has been smoking 1-2 packs of cigarettes per day since the age of 16 (approximately 40 pack years). She admits to "a couple glasses of white wine" every evening after work. She denies use of recreational drugs. NKDA.

D-dimer and lower extremity doppler ultrasonography findings are consistent with deep vein thrombosis.

The patient is hospitalized and treated with subcutaneous low molecular weight heparin (LMWH) (an anticoagulant). In preparation for discharge, she is evaluated for treatment with an oral direct-acting anticoagulant (DOAC) metabolized to inactive metabolites by CYP3A4.

i) What is the likely effect of the concomitant administration carbamazepine and the oral anticoagulant?

- A. Decreased blood levels of carbamazepine
- B. Decreased blood levels of the oral anticoagulant*
- C. Increased blood levels of carbamazepine
- D. Increased blood levels of the oral anticoagulant



Source: Kevin J. Knoop, Lawrence B. Stack, Alan B. Storrow, R. Jason Thurman: The Atlas of Emergency Medicine, 5e Copyright © McGraw Hill. All rights reserved.

ii) Which one of the following options is the most appropriate approach to anticoagulant therapy for this patient?

- A. Discontinue the carbamazepine and prescribe DOAC therapy.
- B. Continue carbamazepine and prescribe a higher dose of the DOAC.
- C. Continue carbamazepine and prescribe self-injections of LMWH.*
- D. Switch the carbamazepine to a different anti-mania drug and prescribe the LMWH.

Definitions:

- Carbamazepine is an antiseizure drug also used in the management mania of bipolar disorders.
- Lithium is an anti-mania drug excreted in the urine (and sweat and saliva) as unchanged drug.
- Heparin does not undergo phase 1 or phase 2 metabolism.
- NKDA: no known drug allergies

Hint: Highlight in the vignette the clues to help you keep track of them and then identify the answers.

Rationale: B. and C.

The patient's bipolar disorder is stable and the medications should not be changed.

LMWH is an anticoagulant that does not interact with the patient's other medications. It is safe and efficacious when used as directed.

The patient is ultimately sent home on subcutaneous LMWH. She is taught how to self-administer the drug.

- The key to the correct answer is determining the *desired beneficial effect* and *minimizing adverse effects* – optimizing drug therapy for the individual patient.

Mechanism of enzyme induction:

- Induction of P450 enzymes results from increased synthesis of the affected enzyme and of the heme, the cofactor. Thus, an increased number of enzymes increases the rate and extent of drug metabolism, which reduces the fraction of the dose that reaches the systemic circulation → reduces the bioavailability of the drug.

Carbamazepine is a strong inducer of CYP3A4 – this drug is starred on the list of inducers.

Why is it so important to memorize the list of enzyme inducers and frequently-used enzyme inhibitors?

This knowledge has significant implications for patient care.

It is Boards relevant.

1) Carbamazepine

- Carbamazepine is an important drug for management of seizure disorders, the mania of bipolar disorders, and trigeminal neuralgia.

- Carbamazepine is frequently used in combination with other drugs.

Carbamazepine has a high potential for pharmacodynamic interactions and metabolism/transport related interactions.

2) DOACs

- Direct oral anticoagulants are convenient, safe, and efficacious when used as directed and don't require frequent monitoring.
- DOACs are CYP3A4 substrates, as mentioned, and frequently used in combination with other drugs.

Inhibition of a DOAC's metabolism increases the risk of major bleeding.

Conclusion: The carbamazepine-oral anticoagulant (DOAC) combination should be avoided.

3) LMWH

- Low molecular weight heparin is relatively convenient (subQ self-injection), safe, and efficacious when used as directed and doesn't require frequent monitoring.
- LMWH has a low potential for metabolism/transport-related drug interactions.

Access Medicine The Atlas of Emergency Medicine, 5e; Chapter 12-18: Deep Vein Thrombosis
Figure 12-45

TEST YOUR KNOWLEDGE OF MECHANISMS AND EFFECTS

Test-taking skills:

The purpose of this exercise is to help you differentiate the information (clues) you need to identify the correct option in case vignette MCQs from the information you do not need.

In the following examples, the name of the drug, action, and therapeutic uses are given. Please identify the mechanism of the drug interaction and potential clinical effect.

YOU DO NOT NEED TO MEMORIZE THE DRUG NAMES AT THIS TIME – JUST THE MECHANISMS OF DRUG INTERACTIONS.

Drug combination	Mechanism, Effect, and Action to mitigate effect
• Omeprazole increases gastric pH for treatment of gastroesophageal reflux • Itraconazole is an antifungal agent that requires an acid pH for absorption.	

· Omeprazole is a gastric acid suppressant and strong CYP2C19 inhibitor. · Clopidogrel is an antiplatelet prodrug that is converted to its active form by CYP2C19.	
· Diazepam is a positive allosteric modulator that potentiates the effects of the neurotransmitter GABA. It is an anxiolytic, sedative, and antiseizure drug. · Alcohol (ethanol) is a recreational beverage and CNS depressant by its action as a GABA-A receptor positive allosteric modulator.	
· Cholestyramine is a resin that is not absorbed from the GI tract and forms a complex with bile acids and many drugs. · Many drugs	
· Naproxen (Aleve) is a nonsteroidal anti-inflammatory drug (NSAID) for pain relief. It reduces platelet function. · Apixaban is a direct oral anticoagulant (DOAC) for treatment and prevention of thromboemboli (clots).	
· Nirmatrelvir-ritonavir (Paxlovid) is an antiviral agent for Covid treatment. Ritonavir is a powerful CYP3A4 inhibitor. · Many drugs are metabolized by CYP3A4.	
· Methotrexate is for the treatment of rheumatoid arthritis (and other autoimmune diseases). Serious AEs include hepatotoxicity and bone marrow toxicity. · Leflunomide is for rheumatoid arthritis by a different mechanism. It is toxic to the liver and bone marrow.	
· Azithromycin is an antibiotic that inhibits P-glycoprotein efflux pump. · Digoxin increases the force of myocardial contraction (cardiotonic). It is a substrate	

of P-glycoprotein and has a narrow therapeutic window (small TI).	
- Propranolol is a nonselective beta blocker for the treatment of hypertension, angina, and certain arrhythmias. It can cause bronchospasm in susceptible patients. - People with asthma have airway hyperresponsiveness – an exaggerated contractile response of the airways to a variety of stimuli.	
- Penicillin is secreted as unchanged drug into the nephron via OAT. Penicillin has a very short duration of action. - Probenecid inhibits OAT	

ANSWERS

- Omeprazole increases gastric pH for treatment of gastroesophageal reflux - Itraconazole is an antifungal agent that requires an acid pH for absorption.	PK: Absorption Itraconazole systemic levels are decreased, which could reduce its antifungal efficacy. <i>Itraconazole capsules should be taken with food and an acidic beverage such as ginger ale or cola.</i> FYI: Itraconazole (capsule) is almost entirely in its cationic form in pH 5 to 9. Proton pump inhibitors raise the gastric pH from 2 to over 6.
- Omeprazole is a gastric acid suppressant and strong CYP2C19 inhibitor. - Clopidogrel is an antiplatelet prodrug that is converted to its active form by CYP2C19.	PK: Metabolic activation Clopidogrel is a prodrug (inactive, by definition). Inhibiting CYP2C19 by would reduce conversion of clopidogrel to the active metabolite. Bioavailability would be reduced, which would likely reduce efficacy.
- Diazepam is a positive allosteric modulator that potentiates the effects of the neurotransmitter GABA. It is an anxiolytic, sedative, and antiseizure drug. - Alcohol (ethanol) is a recreational beverage and CNS depressant by its action as a GABA-A receptor positive allosteric modulator.	PD: Same mechanism → Synergy The benzodiazepine-alcohol combination has synergistic depressant effects on the CNS. The combination in overdose, especially combined with an opioid, can lead to death. <i>Avoid the combination.</i>

- Cholestyramine is a resin that is not absorbed from the GI tract and forms a complex with bile acids and many drugs. - Many drugs	PK: Absorption - Cholestyramine complexes with many drugs, preventing absorption - The complex is excreted in the feces → ↓ efficacy of the drug.
- Naproxen (Aleve) is a nonsteroidal anti-inflammatory drug (NSAID) for pain relief. It reduces platelet function. - Apixaban is a direct oral anticoagulant (DOAC) for treatment and prevention of thromboemboli (clots).	PD: Different mechanisms, different uses, but both increase the risk of bleeding Concurrent use of an NSAID and an anticoagulant increases the risk of bleeding, esp GI bleeding. - Effect is related to patient factors and NSAID dose and duration. <i>Clinical benefit and risk should be assessed in each patient</i> <i>This combination – any combination of drugs that cause bleeding – is contraindicated in patients with active bleeding.</i>
- Nirmatrelvir-ritonavir (Paxlovid) is an antiviral agent for Covid treatment. Ritonavir is a powerful CYP3A4 inhibitor. - Many drugs are metabolized by CYP3A4	PK: Metabolism Inhibiting CYP3A4 enzymes can lead to elevated blood levels of many, many drugs and increase the risk of adverse effects. <i>Refer to published Drug Interactions information for prescribing guidance for the individual patient.</i>
- Methotrexate is for the treatment of rheumatoid arthritis, cancer, and other indications. Serious AEs include hepatotoxicity and bone marrow toxicity. - Leflunomide is for rheumatoid arthritis by a different mechanism. It is toxic to the liver and bone marrow.	PD: Interaction involving same toxicities May be an additive or synergistic effect. Hepatotoxicity and bone marrow toxicity may be enhanced by this drug combination. <i>Monitor for hematologic and hepatotoxicity.</i> <i>If toxicity occurs, discontinue leflunomide and initiate accelerated drug elimination procedure (LEF $t_{1/2}$ 18 days).</i>
- Azithromycin is an antibiotic that inhibits P-glycoprotein efflux pump. - Digoxin increases the force of myocardial contraction (cardiotonic). It is a substrate of P-glycoprotein and has a narrow therapeutic window (small TI).	PK: Distribution / Excretion Azithromycin decreases P-gp efflux of digoxin → increases digoxin absorption and decreases digoxin secretion into the proximal tubule. Elevated serum levels of digoxin can lead to cardiac arrhythmia. <i>Monitor for toxic effects.</i> <i>Consider therapy modification.</i>

- Propranolol is a nonselective beta blocker for the treatment of hypertension, angina, and certain arrhythmias. It can cause bronchospasm in susceptible patients. - People with asthma have airway hyperresponsiveness – an exaggerated contractile response of the airways to a variety of stimuli.	Drug-Disease Interaction Beta blockers, particularly nonselective agents, should be avoided in asthmatic patients because the beta blocker can trigger bronchospasm. <i>Consider therapy modification.</i>
- Penicillin is secreted as unchanged drug into the nephron via OAT. Penicillin has a very short duration of action. - Probenecid inhibits OAT	Probenecid inhibits penicillin excretion, which reduces penicillin's clearance, increases penicillin's blood levels, and its duration of action. <i>This is an example of a beneficial DDI.</i>

SUMMARY

DRUG TOXICITY

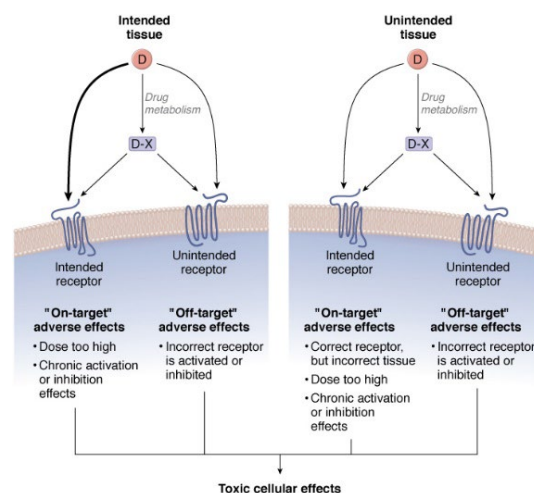
- Adverse drug events (ADEs) are defined as any injuries resulting from medication use, including physical harm, mental harm, or loss of function.
- Medication toxicity is a major cause of hospitalization in older patients.
- Polypharmacy and age-related changes in pharmacokinetics and pharmacodynamics contribute to the risk of ADEs.
- Adverse drug reactions are classified as on-target and off-target, common and predictable, and rare and usually unpredictable.
- Apoptosis, necrosis, mutagenesis and carcinogenesis can result from toxicity of drugs on cells.
- Drug toxicities to cellular and mitochondrial proteins can result in organ toxicity, such as hepatotoxicity, nephrotoxicity, pulmonary toxicity, etc.
- Adverse effects can manifest in minutes to hours (acutely), weeks to months or years after the drug has been discontinued (delayed effects), or cause cumulative effects with chronic drug exposure.
- Effects may be local or systemic.
- The reversibility of toxic effects is related to the capacity of the injured tissue to recover or regenerate.
- Drugs taken by pregnant women may cause fetal death or retardation of growth and development, such as neural tube defects, cleft palate, heart defects, gastrointestinal defects, and others.
- Neonates can manifest abstinence syndrome resulting from maternal use of addictive drugs.

- Drug makers must include in their labeling relevant information about pregnancy testing, contraception, and infertility for health care providers prescribing for females and males of reproductive potential.
- Polymorphisms in metabolic enzymes are thought to underlie individual differences in the **rate** at which drugs are metabolized. Individuals can be phenotypically grouped as poor, intermediate, extensive, or ultra-rapid metabolizers.
- Polymorphisms in genes encoding drug uptake or drug efflux transporters may be other contributors to variability in drug delivery to target sites and, hence, in drug effects.
- Precision medicine: Applying pharmacogenomics information to individual patient care may help clinicians improve drug safety and efficacy.

DRUG INTERACTIONS

🔑 Drug interactions always should be considered when unexpected responses to drugs occur.

- Interactions between drugs alter the PD or PK of one or both interacting compounds.
- **Additive:** when the combined effect of two drugs equals the sum of the effect of each agent given alone and
- **Synergistic/Potentiation:** when the combined effect exceeds the sum of the effects of each drug given alone (supra-additive effects)
- **Antagonism:** the interference of one drug with the action of another.
- **Functional = physiological antagonism:** when two chemicals produce opposite effects on the same physiological function.
- **Chemical antagonism:** a compound that directly binds and forms an inactive complex with another substance (drug or toxin), removing the drug or toxin from circulation.
- **Dispositional antagonism:** the alteration of the disposition of a substance (its absorption, biotransformation, distribution, or excretion) preventing that substance.
- **Receptor antagonism:** blocking the action of the agonist and preventing a cellular response. Receptor in pharmacology is any site of agonist action that produces a biological response.
- Drug interactions occur when one drug modifies the actions or the concentration of another drug in the body.
- Drug interactions can result from pharmacodynamic changes (target and non-receptor interactions), pharmacokinetic alterations, or a combination of both.
- The pharmacodynamic characteristics of different drugs administered concomitantly may lead to additive, synergistic, or antagonistic effects.



- Pharmacokinetic interactions related to absorption, distribution, metabolism, and excretion may increase blood levels (potentially resulting in toxicity) or decrease blood levels (potentially resulting in reduced efficacy).
- Strong inducers of CYP3A4 and P-glycoprotein include carbamazepine, phenobarbital, phenytoin, rifampin, and St. John's wort.
Mnemonic: Precision care promotes robust scientific journeys. Stop smoke
- The CYP1A2 is induced by smoking.
Smokers metabolize some drugs more rapidly than nonsmokers because of CYP1A2 enzyme induction by polycyclic aromatic hydrocarbons, which are formed during incomplete burning of organic substances like tobacco and marijuana.
- Clinically relevant CYP3A4 and P-glycoprotein inhibitors include cimetidine, itraconazole, erythromycin/clarithromycin, ritonavir, grapefruit juice and many others.
Mnemonic: Clinical insights enhance care. Research grows.
- All other CYPs, UDP-glucuronosyltransferase, and various transporters also mediate drug interactions.
- Exaggerated drug effects and other toxicities, as well as reduced efficacy can all lead to adverse drug events.

Patient Safety: Every step in a process has the potential for failure, to varying degrees.

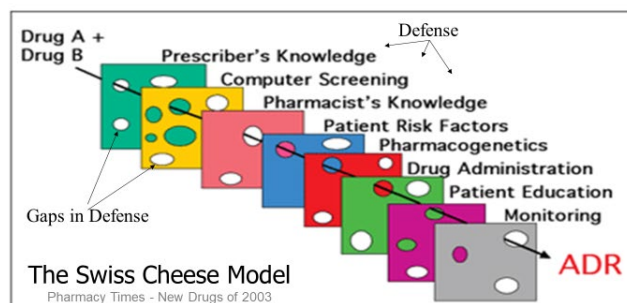


Figure 3—Sometimes “the holes line up,” and the hazard arrow can penetrate each of the defenses unimpeded. Each defense also has other holes, which are called latent failures. These are gaps in the defenses that are not involved in the interaction between Drug A and Drug B, but rather would come into play with other drug interactions. As such, they are accidents waiting to happen. ADR = adverse drug reaction.



Resources:

Access Medicine Fuster & Hurst's The Heart, 15e, 2023; Chapter 36: Atrial Fibrillation and Atrial Flutter

Access Medicine Goodman & Gilman's The Pharmaceutical Basis of Therapeutics, 14e, 2023; Chapter 9: Principles of Clinical Toxicology

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 4: Drug Biotransformation > Enzyme Induction

Access Medicine Katzung's Pharmacology: Examination and Board Review, 14e, 2024; Chapter 4: Drug Metabolism

Access Medicine Review of Medical Microbiology & Immunology: A Guide to Clinical Infectious Diseases, 17e, 2022; Chapter 17: Gram-positive rods; Chapter 65: Hypersensitivity (Allergy) > Immediate (anaphylactic) hypersensitivity. Antigen cross-links IgE on surface of mast cell. Mediators, e.g., histamine, are released from granules in the cytoplasm of the mast cell.

UpToDate Lexidrug monographs: various drug monographs

UpToDate: various topics, literature reviews current through June 2024